Randi C. Williams

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EDUCATION

Massachusetts Institute of Technology (MIT), Cambridge, MA

Doctor of Philosophy in Media, Arts, and Sciences, 5.0/5.0 GPA 2024
Kaufman Teaching Certificate 2022
Dissertation: "Impact.AI: Democratizing AI through K-12 Artificial Intelligence Education"
Committee: Dr. Cynthia Breazeal (advisor, MIT), Dr. Hal Abelson (MIT)
Dr. Tia C. Madkins (UT Austin), Dr. Jean Ryoo (UCLA)

Master of Science in Media, Arts, and Sciences, 5.0/5.0 GPA 2018
Thesis: "PopBots: Leveraging Social Robots to Aid Preschool Children's AI Education"
Committee: Dr. Cynthia Breazeal (advisor, MIT), Dr. Marina Bers (Tufts)
Dr. Paul Harris (Harvard)

University of Maryland, Baltimore County (UMBC), Baltimore, MD

Bachelor of Science in Computer Engineering and Honors College, 3.977/4.0 GPA 2016

PROFESSIONAL EXPERIENCE

CMU Human Computer Interaction Institute, Pittsburgh, PA July 2026
Assistant Research Professor

Day of AI, Cambridge, MA 2025 – 2026
Research Lead, direct report to President Ethan Berman

- Spearhead global research and deployments of K-12 AI education programs, aligning technical curricula with national educational mandates.
- Directed mixed-methods research evaluations to define global frameworks and assessments for AI literacy, directly translating education and HCI theory into student-centered, actionable AI policies.
- Awarded a grant from the Dr. Ibrahim El-Hefni Technical Training Foundation to launch Day of AI Rwanda.

Algorithmic Justice League, Cambridge, MA 2024 – 2025
Program Manager, direct report to President Dr. Joy Buolamwini

- Synthesized complex research on algorithmic bias into accessible public-interest frameworks.
- Drove civic engagement for over 600 stakeholders per quarter in organizational programs.
- Led organizational strategy for AI accountability, turning technical and societal harms research into actionable policy and consumable media narratives.

MIT Media Lab, Personal Robots Group, Cambridge, MA 2016 – 2024
Graduate Research Assistant advised by Prof. Cynthia Breazeal

- Architected open-source interactive robot and AI programming platforms with over 1,000 global users, expanding worldwide access to hands-on AI education.
- Architected a training program for 10,000+ K-12 educators teaching hands-on AI + ethics.

Microsoft Research, Research in Software Engineering (RiSE), Seattle, WA 2021
Graduate Research Intern supervised by Dr. Michał Moskal

- Engineered end-to-end TinyML platform to democratize edge computing, enabling students to deploy and audit machine learning models on low-cost hardware.

MIT Lincoln Laboratories, Informatics and Decision Support Group, Lexington, MA 2016
Graduate Research Intern supervised by Dr. Jason Thornton

- Optimized computer vision architectures for increased system reliability and usability, enhancing the transparency of automated decision-support tools.

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TEACHING EXPERIENCE

MIT Educational Justice Initiative (TEJI) and CSAIL, Cambridge, MA — 2022

Brave Behind Bars Intro to Computer Science taught by Marisa Gaetz, Martin Nisser, and Dr. Emily Harburg

- Co-taught introduction to JavaScript, supported students' projects, led office hours.

MIT Program in Media Arts and Sciences and EECS, Cambridge, MA — 2019 – 2020

Democratizing AI through K-12 AI Education for All taught by Profs. Cynthia Breazeal and Hal Abelson

- Developed and structured core content for new project-based course, targeting both undergrad and graduate students across Wellesley, Harvard, and MIT.
- Facilitated students' learning through project advising, design critiques, and class discussions.
- Directed significant adaptations and managed student support during the onset of COVID-19.

SELECTED PRESENTATIONS AND INVITED TALKS

Industry: TED "Randi Williams: play @ TED" (2026), Future of Work Latvia "Humanity and AI: Navigating the Future" (2025), Ceibal "AI is for Everyone: Reimagining Education for an AI-Powered World" (2025), Google Blockly Developer Summit "Unleashing Creativity: Blocks and Curricula for K-12 AI Education" (2023), ASU+GSV "AI Ethics Curricula in K-8 Classrooms" (2020)

Policy: Association for the Development of Education in Africa (ADEA) "MIT and UNESCO Present: AI and Digital Transformation" (2025), World Bank "Demystifying AI" (2024)

Research: MIT Media Lab "The Future of AI in Education" (2026), Universidade Senai Cimatec "Measuring the Impact of a Global AI Literacy Program" (2026), Northeastern "Teaching kids (and teachers) about AI" (2026), Harvard Graduate School of Education "Inclusive AI Education to Empower Future Generations" (2024), Georgia Tech "Conceptualizing Tools for the Future of AI Education" (2024), Carnegie Mellon University "Developing Tools for the Future of Education with AI" (2024), MIT Generative AI Week "Reinventing the Learner Experience" (2023)

RESEARCH FELLOWSHIPS AND GRANTS

- Dr. Ibrahim El-Hefni Technical Training Foundation — 2025
- Microsoft Research (10 recipients out of 500 applications), PhD Fellowship — 2021
- LEGO Papert Fellowship, MIT Media Lab (3 recipients, nominated by faculty) — 2019
- National Science Foundation, Graduate Research Fellowship — 2018
- MIT Ida Green Fellowship — 2016

HONORS AND AWARDS

- Computer Science Ed Week CS Hero (nominated by CSTA board) — 2023
- Cambridge Science Festival Curious Scientist of the Year (1 recipient, nominated by the city) — 2021
- MIT Graduate Woman of Excellence — 2019
- MIT Unsung Hero (1 recipient, nominated by peers) — 2019

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PEER-REVIEWED PUBLICATIONS

Areas of contribution: AI Education, AI Literacy, Human-Computer Interaction
Google Scholar citations = 3029, h-index: 16, i10-index: 24

Book Chapter

1. Randi Williams. 2026. Using Social Robots to Teach AI to Preschool Children. *Exploring the Potential of Artificial Intelligence in Early Childhood Education: New Learning Experiences for Young Children*. Routledge.

Journals

2. Randi Williams, Safinah Ali, Nisha Devasia, Daniella DiPaola, Jenna Hong, Stephen P. Kaputsos, Brian Jordan, and Cynthia Breazeal. 2022. AI + ethics curricula for middle school youth: lessons learned from three project-based curricula. International Journal of Artificial Intelligence in Education.

Conference Proceedings

3. Raul Alcantara Castillo, Tasneem Burghleh, Deem Alfozan, Randi Williams, Sharifa Alghowinem, Cynthia Breazeal. 2025. Friendelligent: A Collaborative Synchronized Game-Based Platform for AI Education and Autonomous Navigation Curriculum. In Proceedings of the 2025 IEEE Frontiers in Education Conference (FIE '25).
4. Randi Williams*, Safinah Ali*, Raul Alcantara, Tasneem Burghleh, Sharifa Alghowinem, Cynthia Breazeal. 2024. Doodlebot: An Educational Robot for Creativity and AI Literacy. In Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI '24).
5. Randi Williams. 2024. Dr. R.O. Bott Will See You Now: Exploring AI for Wellbeing with Middle School Students. In Proceedings of the 13th Symposium on Education Advances in Artificial Intelligence (EAAI '24). AAAI, Menlo Park, CA, USA.
6. Safinah Ali, Prerna Ravi, Randi Williams, Daniella DiPaola, Cynthia Breazeal. Constructing Dreams Using Generative AI. In Proceedings of the 13th Symposium on Education Advances in Artificial Intelligence (EAAI '24). AAAI, Menlo Park, CA, USA.
7. David Kim, Prerna Ravi, Daeun Yoo, Randi Williams. 2024. App Planner: Utilizing Generative AI in K-12 Mobile App Development Education. In Proceedings of the 16th ACM SIGCHI Interaction Design and Children (IDC) Conference, ACM.
8. Randi Williams. 2022. Constructionism, Ethics, and Creativity: Developing Tools for the Future of Education with AI. In Proceedings of the 2022 IEEE Symposium on Visual Languages and Human-Centric Computing (IEEE VL/HCC '22).
9. Tejal Reddy, Randi Williams, and Cynthia Breazeal. 2022. LevelUp: Automatic assessment of block-based machine learning projects for AI education. In Proceedings of the 2022 IEEE Symposium on Visual Languages and Human-Centric Computing (IEEE VL/HCC '22). * *Best paper award* *
10. Randi Williams, Michal Moskał, and Peli de Halleux. 2022. ML Blocks: A block-based, graphical user interface for creating TinyML models. In Proceedings of the 2022 IEEE Symposium on Visual Languages and Human-Centric Computing (IEEE VL/HCC '22).
11. Tejal Reddy, Randi Williams, Cynthia Breazeal. 2021. Text Classification for AI Education. In Proceedings of the 52nd ACM Technical Symposium on Computer Science Education (SIGCSE'21). * *Won first place in ACM undergraduate student research competition.* *
12. Randi Williams. 2021. How to Train Your Robot: Project-Based AI Education for Middle School Classrooms. In Proceedings of the 52nd ACM Technical Symposium on Computer Science Education (SIGCSE'21).
13. Brian Jordan, Nisha Devasia, Jenna Hong, Randi Williams, Cynthia Breazeal. 2021. PoseBlocks: A Toolkit for Creating (and Dancing) with AI. In Proceedings of the 10th Symposium on Education Advances in Artificial Intelligence (EAAI '21). AAAI, Menlo Park, CA, USA. *Supervised graduate/undergraduate team who took MAS.S65 course.*
14. Randi Williams, Stephen P. Kaputsos, Cynthia Breazeal. 2021. Teacher Perspectives on How to Train Your Robot, A Middle School AI and Ethics Curriculum. In Proceedings of the 10th Symposium on Education Advances in Artificial Intelligence (EAAI '21). AAAI, Menlo Park, CA, USA.

15. Phoebe Lin, Jessica Van Brummelen, Galit Lukin, Randi Williams, Cynthia Breazeal. 2020. Zhorai: Designing A Conversational Agent for Children to Explore Machine Learning Concepts. In Proceedings of the 9th Symposium on Education Advances in Artificial Intelligence (EAAI ’20). AAAI, Menlo Park, CA, USA. *Supervised graduate/undergraduate team who took MAS.S65 course.*
16. Randi Williams, Hae Won Park, and Cynthia Breazeal. 2019. A is for Artificial Intelligence. In Proceedings of the 2019 Conference on Human Factors in Computing (CHI ’19). ACM, New York, USA.
17. Randi Williams, Hae Won Park, Lauren Oh, and Cynthia Breazeal. 2019. PopBots: Designing an Artificial Intelligence Curriculum for Early Childhood Education. In Proceedings of the 9th Symposium on Education Advances in Artificial Intelligence (EAAI ’19). AAAI, Menlo Park, CA, USA.
18. Jacqueline M. Kory-Westlund, J. M., Hae Won Park, Randi Williams, and Cynthia Breazeal. 2018. Measuring Young Children’s Long-term Relationships with Social Robots. In Proceedings of the 17th ACM Interaction Design and Children Conference (IDC) (pp. 207-218). ACM: New York, NY.
19. Stefania Druga, Randi Williams, Hae Won Park, and Cynthia Breazeal. 2018. How smart are the smart toys?: children and parents’ agent interaction and intelligence attribution. In Proceedings of the 17th ACM Conference on Interaction Design and Children (IDC ’18). ACM, New York, NY, USA, 231-240. DOI: https://doi.org/10.1145/3202185.3202741.
20. Randi Williams, Christian Vázquez Machado, Stefania Druga, Cynthia Breazeal, and Pattie Maes. 2018. “My doll says it’s ok”: a study of children’s conformity to a talking doll. In Proceedings of the 17th ACM Conference on Interaction Design and Children (IDC ’18). ACM, New York, NY, USA, 625-631. DOI: https://doi.org/10.1145/3202185.3210788.
21. Randi Williams*, Stefania Druga*, Mitch Resnick, and Cynthia Breazeal. 2017. “Hey Google, is it OK if I Eat You?”: Initial Explorations in Child-Agent Interaction. In Proceedings of the 16th ACM SIGCHI Interaction Design and Children (IDC) Conference, ACM.

PROFESSIONAL SERVICE

- Black in Robotics, Boston Chapter Co-director — 2021 – 2024
- Black in AI Research (BlackAIR) Summer Research Mentor — 2021
- MIT Graduate Student Council, Diversity Conduit — 2018 – 2021
- MIT Media Lab Culture Working Group, Member — 2019 – 2020
- IBM Watson AI XPRIZE Judge — 2019 – 2020
- MIT Graduate Student of Color Advisory Council, Member — 2018 – 2020